YOUTUBE TRENDING
Data Lakehouse

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Data Science & Big Data Engineering Project

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1. Project overview

This project focuses on analyzing Youtube trending data to demonstrate core concepts of big data ingestion, cleaning, and analysis using Snowflake SQL. The dataset is hosted in Azure Blob Storage and included two key components:

- Trending data (CSV): Daily records of trending Youtube videos by country, containing metrics such as views, likes, dislikes, and comments.
- Category data (JSON): Metadata mapping each video's categoryId to a descriptive category_title.

2. Data Ingestion

Create a working database

We create a working database name assignment_1 to isolate all objects for the assignment, avoids name collisions, and makes marking reproducible.

Create a secure stage to Azure

Snowflake is then connected to this storage using a secure stage 'stage_assignment' with a SAS token, which ensures temporary, scopes access to the container without exposing sensitive credentials.

Define file format

File formats are defined for each data type to handle parsing correctly. For the CSV trending data, the format includes specifications to skip header rows, handle fields enclosed by quotation marks, and standardise common null tokens into proper SQL NULL values. This design choice ensures robust ingestion, particularly for titles containing commas or inconsistent null markers. A JSON file format is also created for the category data, allowing Snowflake to interpret semi-structured content.

Create external table

External tables are built directly over the files in Azure, using file patterns to restrict the scope to the intended datasets. This approach allows Snowflake to read the files in place without creating copies. A refresh command is issued to register the latest files, and small sample queries (SELECT … LIMIT 1) are used to verify connectivity and schema readability.

Create internal table

For the trending data, the table table_youtube_trending is created with explicit casts for each column (e.g., integers for view counts and dates for trending dates). Similarly, the category metadata is materialized into the table_youtube_category table using the LATERAL FLATTEN function to expand the nested items array in the JSON structure. The country field is derived from the file name to support country-aware joins.

Create final table

table_youtube_final is created by joining the two materialized tables on both country and categoryId. To provide a stable, unique identifier for each row, a key is generated using the UUID_STRING() function. This is an important step in enabling safe row-level operations such as de-duplication in subsequent tasks.

The resulting table_youtube_final contains 2,667,041 rows, consistent with the assignment brief.

Results (just now)

Table Chart 2,667,041 rows 3.4s

	ID	VIDEO_ID	TITLE	PUBLISHEDAT	CHANNELID
1	53d10772-1c61-45b0-bdd9-1d445ae2f3	oHpgu-cTjyM	Where Gaming Begins: Ep. 2 | AMD Radeon™ RX 6000 Serie	2020-10-28 16:00:09.000 +0000	UCHQDjDDW8w2RieO-luqYlyg
2	aa5a92ec-273d-408d-85dc-b1ac281085	QtMzV73NAgk	PlayStation 5 Unboxing & Accessories!	2020-10-27 12:00:37.000 +0000	UCBJycsmduvYEL83R_U4JriQ
3	fffa8ada-aaed-47b8-88e8-ef5587e840b5	qWkJoo7D-MI	Our unpopular opinions!! *don't hate us*	2020-10-28 17:00:11.000 +0000	UCFanrVWRodCwCw43U7KBAQg
4	cc95553d-5201-4ade-b6b9-2d59ba2f4c	4qyWLVWg7V4	How to Carve a Simple and Scary Pumpkin Face 2	2020-10-28 13:30:00.000 +0000	UCSC1HqVmTaE4Shn32ihbC7w
5	91b105b0-1c78-4a50-a3db-c3f3314001	mDG4R2tjKKA	what i eat in a day	2020-10-27 17:00:24.000 +0000	UC9PodfZydKm8akexKRn_2bw
6	88a0e340-bd73-4d31-9e37-9b0989d5a	a5t0PLqMy74	World Premiere: BUGATTI Bolide	2020-10-28 11:00:15.000 +0000	UC2qmmoQdJXLmre794U6zjXw
7	990173bc-e2dd-4de2-946e-6f4553690c	TFzkbos0oeo	You Really Got Me | Cinematic Trailer - League of Legends:	2020-10-28 05:30:00.000 +0000	UCd0ZomI5mPkrV735wul5kiw
8	84cffa3d-f966-4521-b7b0-f63ddedeba1	01H06TbMs4g	Tamar Braxton Breaks Her Silence Following Suicide Attemp	2020-10-28 22:04:40.000 +0000	UCcq6gCTW4ntQny2ACWXXLlw
9	1b168fb9-c3ed-45a0-adc9-3b811ff5198	wP9TzdeLHvc	CL +HWA Dance Performance Video+	2020-10-29 04:00:11.000 +0000	UCuX8lXlexG8dRXFx54CVm2Q
10	922ad86f-32f3-4309-a556-50af113d194	oLZUDfT4iPs	NVIDIA Never saw this coming... AMD on top??	2020-10-28 18:10:42.000 +0000	UCkWQ0gDrqOCarmUKmppD7GG

Figure 1: The table_youtube_final has 2,667,041 rows

3. Data Cleaning

Once the unified dataset table_youtube_final was created, this stage involves identifying and resolving data quality issues.

The first issue examined is the consistency of category information. By grouping records in the table_youtube_category table by category_title, duplicates are identified that ignore categoryId.

Results (just now)

Table Chart 30 rows 2.6s

	CATEGORY_TITLE	COUNT
1	Action/Adventure	10
2	Anime/Animation	10
3	Autos & Vehicles	10
4	Classics	10
5	Comedy	20
6	Documentary	10
7	Drama	10
8	Education	10
9	Entertainment	10
10	Family	10

Figure 2: Results of category_title having duplicates if ignore categoryid

Another query identifies category 'Nonprofits & Activism' that is unique to a single country (US). Both checks are necessary to understand the distribution of category metadata and to confirm that joining on categoryId is more reliable than joining on category names.

Results (just now)

Table Chart 1 row 1.3s

	CATEGORY_TITLE	COUNTRY	COUNT_COUNTRY
1	Nonprofits & Activism	US	1

Figure 3: 'Nonprofits & Activism' is the category_title only appears in one country

We also identify that there are 1,563 missing category_title values in the table_youtube_final table, then replace them with the 'Nonprofits & Activism' category.

Results (just now)

Table Chart 1 row 407ms

	CATEGORYID	COUNT(*)
1	29	1563

Figure 4: Categoryid '29' has missing category_title

In addition to category inconsistencies, checks are also run for invalid or incomplete video information. A query identifies there are two distinct videos that lacked a channelTitle, returning only the video titles and IDs.

Results (3 minutes ago)

Table Chart 2 rows 285ms

	TITLE	VIDEO_ID
1	Kala Official Teaser | Tovino Thomas | Rohith V S | Juvis Productions | Adventure Company	9b9MovPPewk
2	APRÈS 1 AN, JE PEUX ENFIN VOUS L'ANNONCER !	UuLTojgml_o

Figure 5: 2 distinct video do not have channeltitle

Another query is done to check, and 32,081 records where the video_id contains the token '#NAME?' are deleted, which is an error marker rather than a valid identifier.

Results (3 minutes ago)

Table Chart 1 row 3.2s

	# number of rows deleted
1	32081

Figure 6: There are 32,081 video_id containing the token '#NAME?'

Duplicate records are another key challenge. Since the dataset could contain multiple rows with the same video_id, country, and trending_date but different metrics, a window function with ROW_NUMBER() is used to rank duplicates within each partition by view_count. The assumption here was that the row with the highest view count represented the most accurate version of the trending record, while the others were redundant.

All lower-ranked duplicates (37,466 rows) are inserted into a separate table, table_youtube_duplicates, both for auditability and to allow safe deletion from the main table.

Results (just now)

Table Chart 37,466 rows 654ms

	ID	VIDEO_ID	TITLE	PUBLISHEDAT	CHANNELID	C
1	9f99b283-bd84-4906-9f73-f634eb0279	94JBhbeTiMI	I GOT MARRIED TO MY BEST FRIEND FOR 24 HOURS!!	2021-02-28 01:30:01.000 +0000	UC56D-IHcUvLVFTX_8NpQMXg	Brent
2	53247d71-a8b3-460e-9c2a-4a5d5fa05	k97i3kalD8g	Roman Reigns Brutally Attacks Dominik and Rey Mysterio	2021-06-05 11:45:01.000 +0000	UC_WKb6N9iTGc77hxwXLDrbA	Sony
3	9c3b1415-7a6e-4f06-a5a2-400999359	bYa11TrLn5U	Se oye como las olas del mar, así es el extraño socavón q	2021-05-31 13:25:38.000 +0000	UCW7vke0bldunYJzPEYGW-aQ	El So
4	3656432f-2fb4-4400-97b3-e2560f6a5	Us3R58tbebc	【元プロ野球選手コラボ】里崎さんに相談したら神アドバイ	2021-06-01 11:00:20.000 +0000	UCDLqvTPnN70jKZNqaDF4GwQ	友ch
5	5b41fe49-3b9e-4abe-86ab-61d923cf3c	wvIPYXXFTuE	Selena Gomez & Rauw Alejandro - Baila Conmigo (Premio	2021-02-19 07:30:03.000 +0000	UCjK8ORC71kwyj1DWFwril_A	Seler
6	65a811a5-6f17-4e98-aca0-d4bddb1e4	ymt0Ay1kLlo	Adham Nabulsi - Khayef (Official Music Video) | أدهم نابلسي - خايف	2021-02-21 15:30:05.000 +0000	UCPibKXYTX0Yu2578k07EOhg	Adha
7	867c6728-4703-46a0-956f-829554797	ASzOzrB-a9E	Battlefield 2042 Official Reveal Trailer (ft. 2WEI)	2021-06-09 15:00:14.000 +0000	UCvNBXWGykQrWb7kPAn5eLUQ	Battl
8	c6e6cab5-d5b1-4b9b-b8fe-81b784f467	K-pB9emi41U	Minecraft But YouTubers are Totems...	2021-05-31 14:15:01.000 +0000	UCJZam2u1G0syq3kyqrCXrNw	Prest
9	4921c0b4-c4a9-427f-b12a-eb5588360	RpNwY-_q4rY	기억상실 나천재 [총몇명 스토리2]	2021-06-02 09:10:40.000 +0000	UCRuSxVu4iqTK5kCh90ntAgA	총몇명
10	d29ecc2c-bc7b-47ec-b4f8-51b674ce88	fbWAQpokV0w	LES OVNIS SONT DISPONIBLES SUR FORTNITE !	2021-06-02 04:03:00.000 +0000	UCHlg01d4juXgHgGOOQIGY9A	Thoo

Figure 7: There are 37,466 duplicate rows in table_youtube_duplicates

A subsequent delete operation removes these duplicates from table_youtube_final, using the unique surrogate key (id). After completing the cleaning operations, the resulting total is 2,597,494 rows, which matches the expected figure specified in the assignment brief.

Results (5 minutes ago)

Table | Chart | 2,597,494 rows | 3.4s

	ID	VIDEO_ID	TITLE	PUBLISHEDAT	CHANNELID
1	7857ff78-4758-45ed-b9f7-8fa34b7e85	UQxUZOnqkTk	Wie sollte man 1.000€, 10.000€, 100.000€, 1 Mio. € anleg	2022-08-21 08:00:17.000 +0000	UCeARcCUiZg79SQQ-2_XNIXQ
2	a6d5867b-7a58-48ac-b3fa-f44c01f222	OTd28lXLEfc	Siilawy - لما تكوني (Official Music Video)	2022-08-19 15:00:12.000 +0000	UCJO1mAZZyib8bYHAlNFjSBQ
3	089087ed-b384-46dd-974f-84c30a44	lfeXlKWF-34	Minecraft, Aber Äpfel Sind Super OP	2022-08-20 10:15:01.000 +0000	UC4UDgtqBSQZ4yk7H_sEiFWA
4	a07061f7-03af-4b64-8e00-5d52f84a3f	n3PX--PaxGY	Epischen Bogen bauen aus Holz!	2022-08-19 16:58:52.000 +0000	UCzidBg7JPR53EP5B5×65jpA
5	5bdd9ec3-031e-4e37-9c1d-d2db1c63	V61_GE28uF0	Anthony Joshua makes passionate speech & vents frustrati	2022-08-21 00:25:46.000 +0000	UC_JQGBtA7P0RwkRxd7xpJcA
6	cdec6684-d610-48ec-a1d8-b551de6f6	fwOrR_2U7ss	Übernachtung auf Schiffswrack - Durchkreuzt das Gewitter	2022-08-20 08:00:14.000 +0000	UCTWiyoWvQ6CnnRXyWL2mIAA
7	fb0e1617-f331-47a2-9a53-17c0c846e	3uXOcAgMnZI	DER BIWAK SACK war KEINE GUTE IDEE! 😂 7 VS WILD TRA	2022-08-18 13:09:59.000 +0000	UCtckLvYloMnsey8E5qTwWwA
8	bb685a29-997a-4eab-81c6-8f73a2c47	32ADdEL7_XU	Wegen Cristiano Ronaldo: Nagelsmann bringt alle zum Lach	2022-08-19 13:05:57.000 +0000	UClk-KrB_r6b05FrWlQpbV3Q
9	5923a1b2-bb27-42a1-9f98-72b49703	QrbNZVKC19Y	Peter Junker Forst- und Flurwegebau im Einsatz | Fendt Tra	2022-08-19 16:30:06.000 +0000	UClEU2iqj54e-2GgsC5HZUIQ
10	30482b88-ecb7-40c3-9250-ea588aca7	QUZShSYx4NM	Unglaublich, aber legal! Porsche 911 GT3 RS I auto motor u	2022-08-20 07:00:08.000 +0000	UCLINPbYQ9sy6qc-TqtBeVnw

Figure 8: The final table ‘table_youtube_final’ has 2,597,494 rows

4. Data Analysis

The first query identifies the top three Gaming videos per country on 1 April 2024. Using ROW_NUMBER() partitioned by country and ordered by views, the analysis showes which gaming videos are most popular on that date:

	COUNTRY	TITLE	CHANNELTITLE	VIEW_COUNT	RK
1	BR	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1
2	BR	IShowSpeed x MC Kevin O Chris - Amar de (Official Music Video)	IShowSpeed	2971782	2
3	BR	Confrontation - The Skibidi Saga 05	Maxedy	2323375	3
4	CA	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1
5	CA	If my viewers break my secret rule, I ban them	DougDoug	2988844	2
6	CA	Confrontation - The Skibidi Saga 05	Maxedy	2323375	3
7	DE	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1
8	DE	If my viewers break my secret rule, I ban them	DougDoug	2988844	2
9	DE	Season 3 Warzone Launch Trailer - Rebirth Island | Call of Duty: Warzone	Call of Duty	2311131	3
10	FR	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1

Figure 9: Top three Gaming videos per country on 2024-04-01

The second query measures the popularity of BTS by counting distinct video IDs with “BTS” in the title. The results show that KR leads with 468 distinct BTS videos, while IN and the US follow with 288 and 268 respectively. Other countries like CA, MX, JP, and DE also rank highly, confirming BTS’s influence across Asia, North America, and Europe.

	COUNTRY	CT
1	KR	468
2	IN	288
3	US	268
4	CA	262
5	MX	254
6	JP	251
7	DE	242
8	GB	223
9	BR	186
10	FR	167

Figure 10: The number of distinct video with 'BTS' in the title in each country

The third query identifies the most-viewed video for each country and month in 2024 and calculates its likes ratio (likes ÷ views × 100, truncated to two decimals). Results show two clear trends in January 2024. MrBeast leads in Brazil, India, Japan, Korea, and Mexico with his high-stakes challenge videos, while League of Legends' "Still Here" cinematic topes in Germany, France, and the UK. In the US, the GTA VI Trailer dominates with the highest views and a strong likes ratio of 6.72%, reflecting huge anticipation.

	COUNTRY	YEAR_MONTH	TITLE	CHANNELTITLE	CATEGORY_TITLE	VIEW_COUNT	LIKES_RATIO
1	BR	2024-01-01	Survive 100 Days Trapped, Win $500,000	MrBeast	Entertainment	139504939	3.20
2	CA	2024-01-01	Still Here | Season 2024 Cinematic - League o	League of Legends	Gaming	104159411	1.68
3	DE	2024-01-01	Still Here | Season 2024 Cinematic - League o	League of Legends	Gaming	104159411	1.68
4	FR	2024-01-01	Still Here | Season 2024 Cinematic - League o	League of Legends	Gaming	104159411	1.68
5	GB	2024-01-01	Still Here | Season 2024 Cinematic - League o	League of Legends	Gaming	104159411	1.68
6	IN	2024-01-01	Protect $500,000 Keep It!	MrBeast	Entertainment	85458562	4.21
7	JP	2024-01-01	Survive 100 Days Trapped, Win $500,000	MrBeast	Entertainment	137639799	3.22
8	KR	2024-01-01	Survive 100 Days Trapped, Win $500,000	MrBeast	Entertainment	143955997	3.15
9	MX	2024-01-01	Survive 100 Days Trapped, Win $500,000	MrBeast	Entertainment	137639799	3.22
10	US	2024-01-01	Grand Theft Auto VI Trailer 1	Rockstar Games	Gaming	166323421	6.72

Figure 11: The most viewed video and its likes_ratio in each country

At the fourth query, distinct videos are aggregated by category and country to determine which category contributes the largest share of unique content from 2022. Entertainment dominates most countries like India with 42.35%, Korea (33.75%), and Japan (32.09%). Other countries such as Brazil (22.79%), Germany (25.09%), France (22.97%), GB (20.25%), and Mexico (23.92%) also shows Entertainment as the leading category. By contrast, Gaming is the strongest driver of distinct trending videos in North America, such as Canada (21.36%) and US (21.61%).

	COUNTRY	CATEGORY_TITLE	TOTAL_CATEGORY_VIDEO	TOTAL_COUNTRY_VIDEO	PERCENTAGE
1	BR	Entertainment	5417	23760	22.79
2	DE	Entertainment	7709	30719	25.09
3	FR	Entertainment	7548	32849	22.97
4	GB	Entertainment	5643	27855	20.25
5	IN	Entertainment	21281	50250	42.35
6	JP	Entertainment	5658	17627	32.09
7	KR	Entertainment	5122	15175	33.75
8	MX	Entertainment	4195	17532	23.92
9	CA	Gaming	6594	30869	21.36
10	US	Gaming	6226	28799	21.61

Figure 12: The category_title has the most distinct videos and its percentage out of the total distinct number of videos of each country

Finally, channels are ranked by the number of distinct videos produced. Vijay Television leads with 2,049 distinct videos**,** making it the most prolific content producer in the dataset.

	CHANNELTITLE	NUM_VIDEOS
1	Vijay Television	2049

Figure 13: The channeltitle 'Vijay Television' has produced the most distinct videos

5. Business Question

To answer the question rigorously, we first exclude Music and Entertainment as required, then built a record for each (country, category_title, video_id) using a window rank ordered by likes (DESC), views (DESC), latest trending_date. This avoids over-counting the same video across multiple trending days and prioritises high-engagement content. From that view, we count distinct videos per (country, category), identify each country's top category, compute that category's share of the country's distinct videos, and compare those local winners to the global winner to measure alignment and list exceptions. We repeat the analysis for all years and for 2024 only.

At the global level, the conclusion is stable: after removing Music/Entertainment, Sports is the best single bet, with 43,323 distinct trending videos across all years and 3,394 in 2024. This indicates that sports content has the broadest creator footprint and the most consistent ability to trend worldwide among the remaining categories.

	SECTION	RECOMMENDED_CATEGORY	DISTINCT_VIDEOS_WORLDWIDE
1	GLOBAL_RECOMMENDATION_ALL	Sports	43323

Figure 14: Sports has 43,323 distinct videos across all years

	SECTION	RECOMMENDED_CATEGORY	DISTINCT_VIDEOS_WORLDWIDE
1	GLOBAL_RECOMMENDATION_2024	Sports	3394

Figure 15: Sports has 3,394 distinct videos in 2024

However, the country view shows meaningful variation. Across all years, Sports leads in Brazil (32.51%), Germany (23.37%), and the UK (30.43%), but Gaming dominates in Canada (29.60%), France (24.28%), Japan (25.54%), Mexico (27.29%), and the United States (30.84%). Meanwhile, People & Blogs is the top category in India (34.40%) and South Korea (28.76%).

	SECTION	COUNTRY	TOP_CATEGORY_IN_COUNTRY	DISTINCT_TRENDING_VIDEOS	PCT_OF_COUNTRY
1	BEST_BY_COUNTRY_ALL	BR	Sports	7122	32.51
2	BEST_BY_COUNTRY_ALL	CA	Gaming	9913	29.60
3	BEST_BY_COUNTRY_ALL	DE	Sports	7567	23.37
4	BEST_BY_COUNTRY_ALL	FR	Gaming	7788	24.28
5	BEST_BY_COUNTRY_ALL	GB	Sports	9590	30.43
6	BEST_BY_COUNTRY_ALL	IN	People & Blogs	12580	34.40
7	BEST_BY_COUNTRY_ALL	JP	Gaming	4390	25.54
8	BEST_BY_COUNTRY_ALL	KR	People & Blogs	4379	28.76
9	BEST_BY_COUNTRY_ALL	MX	Gaming	4743	27.29
10	BEST_BY_COUNTRY_ALL	US	Gaming	9333	30.84

Figure 16: The top category in each country across all years

In 2024, Sports remains No.1 in Brazil (39.83%), Germany (26.34%), France (27.12%), and the UK (29.34%); Gaming tops Canada (27.89%), Japan (30.49%), Mexico (26.68%), and the United States (29.70%); and People & Blogs continues to lead in India (37.80%) and South Korea (31.91%). These patterns are consistent with the highly engaged examples surfaced by the ranking (e.g., GTA VI Trailer in Gaming and strongly liked People & Blogs uploads).

	SECTION	COUNTRY	TOP_CATEGORY_IN_COUNTRY	DISTINCT_TRENDING_VIDEOS	PCT_OF_COUNTRY
1	BEST_BY_COUNTRY_2024	BR	Sports	668	39.83
2	BEST_BY_COUNTRY_2024	CA	Gaming	629	27.89
3	BEST_BY_COUNTRY_2024	DE	Sports	570	26.34
4	BEST_BY_COUNTRY_2024	FR	Sports	614	27.12
5	BEST_BY_COUNTRY_2024	GB	Sports	636	29.34
6	BEST_BY_COUNTRY_2024	IN	People & Blogs	1052	37.80
7	BEST_BY_COUNTRY_2024	JP	Gaming	337	30.49
8	BEST_BY_COUNTRY_2024	KR	People & Blogs	375	31.91
9	BEST_BY_COUNTRY_2024	MX	Gaming	266	26.68
10	BEST_BY_COUNTRY_2024	US	Gaming	627	29.70

Figure 17: The top category in each country in 2024

Quantitatively, a single global choice does not fit most markets. Using Sports as the global pick matches the country-level winner in only 3 of 10 countries (30.00%) across all years, rising to 4 of 10 (40.00%) in 2024. In other words, a “one-size-fits-all” approach would underperform in most countries.

	SECTION	COUNTRIES_ALIGNED	TOTAL_COUNTRIES	PCT_COUNTRIES_ALIGNED
1	ALIGNMENT_SUMMARY_ALL	3	10	30.00

Figure 18: The alignment of Sports with all countries across all years

	SECTION	COUNTRIES_ALIGNED	TOTAL_COUNTRIES	PCT_COUNTRIES_ALIGNED
1	ALIGNMENT_SUMMARY_2024	4	10	40.00

Figure 19: The alignment of Sports with all countries in 2024

Recommendation: use a two-tier strategy. Globally, prioritise Sports to maximise overall trending potential once Music/Entertainment are removed. Locally, switch to the country's empirically best category: Gaming in the US, Canada, Mexico, and Japan, and People & Blogs in India and South Korea. Within each target category, mirror formats that historically drive high likes and high views, as indicated by the canonical ranking. This "global default + local override" plan captures Sports' broad upside while exploiting country-specific strengths evidenced by distinct-video shares.

6. Issues while coding

An issue we encountered was the inconsistency in how the country field was derived across the two versions of the Part 1 SQL script. In the correct version, the country was extracted directly from the file name using the SPLIT_PART function, which correctly captured all country codes. In the wrong version, however, a long CASE statement was used to manually map only a limited set of countries (e.g., BR, CA, DE, FR, GB, IN, JP, KR, MX, US). Any country not included in this list was assigned as NULL. This led to many rows being stored with missing country values, which directly affected later joins and the way duplicates were grouped in Part 2.

```
-- Create a table named table_youtube_trending
create or replace table table_youtube_trending as
SELECT
    value:c1::varchar AS video_id,
    value:c2::varchar AS title,
    value:c3::timestamp_tz AS publishedAt,
    value:c4::varchar AS channelId,
    value:c5::varchar AS channelTitle,
    value:c6::varchar AS categoryId,
    value:c7::date AS trending_date,
    value:c8::int AS view_count,
    value:c9::int AS likes,
    value:c10::int AS dislikes,
    value:c11::int AS comment_count,
  (CASE
        WHEN METADATA$FILENAME ILIKE '%BR_%' THEN 'BR'
        WHEN METADATA$FILENAME ILIKE '%CA_%' THEN 'CA'
        WHEN METADATA$FILENAME ILIKE '%DE_%' THEN 'DE'
        WHEN METADATA$FILENAME ILIKE '%FR_%' THEN 'FR'
        WHEN METADATA$FILENAME ILIKE '%GB_%' THEN 'GB'
        WHEN METADATA$FILENAME ILIKE '%IN_%' THEN 'IN'
        WHEN METADATA$FILENAME ILIKE '%JP_%' THEN 'JP'
        WHEN METADATA$FILENAME ILIKE '%KR_%' THEN 'KR'
        WHEN METADATA$FILENAME ILIKE '%MX_%' THEN 'MX'
        WHEN METADATA$FILENAME ILIKE '%US_%' THEN 'US'
        ELSE NULL
    END) as COUNTRY
from ex_table_youtube_trending;
select*from table_youtube_trending;
```

Figure 20: The wrong code used to extract countries when creating table_youtube_trending

In addition, the category table creation was inconsistent: one script extracted the country using SPLIT_PART, while the wrong used REGEXP_SUBSTR. Because the trending table and the category table were not aligned in their country extraction methods, some rows failed to match correctly during the join. These mismatches resulted in discrepancies in the final dataset and ultimately changed the number of duplicates detected (87,856 rows). By standardizing the country extraction approach across both tables, we were able to resolve these inconsistencies and confirm the correct duplicate count of 37,466.

```
-- Create a table named table_youtube_category
CREATE OR REPLACE TABLE table_youtube_category AS
SELECT
    UPPER(REGEXP_SUBSTR(metadata$filename, '^(..)'))            AS country,
    f.value:id::NUMBER                                         AS categoryId,
    f.value:snippet.title::STRING                              AS category_title
FROM ex_table_youtube_category t,
     LATERAL FLATTEN(input => t.value:items) f;
select*from table_youtube_category;
```

Figure 21: The wrong code used to extract countries when creating table_youtube_category

Table | Chart | 87,856 rows | 38ms

	ID	VIDEO_ID	TITLE	PUBLISHEDAT	CHANNELID	CHANN
1	7ed2fcc0-4c63-443d-88cf-6a2765417e	vxmOKVRJPks	ESSE É MEU CARRO NOVO ! *NISSAN GTR*	2021-05-29 14:00:31.000 +0000	UChMJkHCD2wdvSO7CMOuLR-w	Renato Ga
2	ecf605fe-3141-4e68-94f3-eb465d3eec	BC19kwABFwc	Dua Lipa - Love Again (Official Music Video)	2021-06-04 12:00:11.000 +0000	UC-J-KZfRV8c13fOCkhXdLiQ	Dua Lipa
3	d1ec7576-321a-497b-aeaf-bbdf64e02f	xl7wBpOU9YU	HIGHLIGHTS | Canada v Suriname - CONCACAF	2021-06-09 03:25:18.000 +0000	UCC8KIJfNKi_e6TC6RH8GDIA	OneSocce
4	65f13afb-0c7c-4737-97c3-9e5d6016cc	Q2Yyb5OWdYg	Cosas ÚNICAS de Minecraft BEDROCK	2021-06-02 22:00:09.000 +0000	UCTZ_HMjMNDi8Jn3rDWgcaRg	Gelo
5	b1dbc792-0aef-48c4-97a7-6c7b5350fd	1WEAJ-DFkHE	$1 vs $500,000 Plane Ticket!	2023-04-01 20:00:04.000 +0000	UCX6OQ3DkcsbYNE6H8uQQuVA	MrBeast
6	1e336ab8-215d-4832-ae49-ac4fafba10	1WEAJ-DFkHE	$1 vs $500,000 Plane Ticket!	2023-04-01 20:00:04.000 +0000	UCX6OQ3DkcsbYNE6H8uQQuVA	MrBeast
7	b67f3eec-70e6-48b5-89e3-31bc6de1f5	YDehiCsDar8	WHEN YOU OWE AN AFRICAN PARENT MONEY	2021-06-07 17:06:14.000 +0000	UCUocOQtoaHPuT0K8IME559A	SamSpedy
8	997919a8-03b0-4a0d-ac8c-a8824ccaf6	5wq8kkdDDts	[최초공개] ENHYPEN(엔하이픈) - Drunk-Dazed (4K	2021-04-26 12:08:02.000 +0000	UCTQVlXvcHrR9jYoJ6qaBAow	M2
9	72da9613-21f1-4e0a-8d64-1a1268bf32	9z8sEadieXA	Race Highlights | 2024 Australian Grand Prix	2024-03-24 06:25:42.000 +0000	UCB_qr75-ydFVKSF9Dmo6izg	FORMULA
10	275c118c-7327-421d-af9c-1ccf9ebe59	9z8sEadieXA	Race Highlights | 2024 Australian Grand Prix	2024-03-24 06:25:42.000 +0000	UCB_qr75-ydFVKSF9Dmo6izg	FORMULA

Figure 22: The wrong table_youtube_duplicates with 87,856 rows

7. Conclusion

Throughout this project, we explore the process of ingesting, cleaning, and integrating YouTube trending data into Snowflake, with a particular focus on identifying and resolving duplicate records. The main challenge stems from inconsistencies between two versions of the Part 1 SQL scripts, especially in the way the country field and category *table* derive. These mismatches initially produce different results in Part 2 and lead to confusion around the correct number of duplicates. By carefully comparing both scripts and standardizing the approach to extracting country codes and aligning category joins, we resolve the discrepancies. The outcome confirms that 2,597,494 rows in the final dataset, which is consistent and reliable after applying the corrected logic.

Excluding Music and Entertainment, Sports is our global launch category by breadth (43,323 distinct trending videos across all years; 3,394 in 2024). However, a single playbook will not work everywhere: the global pick matches the local top 1 in only 30% of countries across all years and 40% in 2024. Several markets consistently prefer other categories — Gaming in the US/CA/MX/JP (and FR on the all-years view) and People & Blogs in IN/KR—mirrored by their top-liked videos (e.g., GTA VI Trailer, BABYMONSTER "SHEESH"). Therefore, our recommended strategy is two-tiered**:** use Sports as the global default to maximize overall discoverability, and apply local overrides**—**Gaming in US/CA/MX/JP and People & Blogs in IN/KR — timed to regional release cycles and executed with short, high-engagement formats. This balances global scale with country-level fit to maximize chances of trending.